
UNIT 4 BIOMETRICS*

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LEARNING OBJECTIVES

After the completion of this unit you will be able to understand:

- biometrics, its scope and role in forensic anthropology;
- the use of human characteristics in personal identification with the help of computer technology; and
- characteristics and the methodology used in biometrics for the individual identification.

4.0 INTRODUCTION

‘Forensics’ means *‘for the court’* and it relies on the evidence and identification. You are well aware that forensic analysis is dependent on the evidence collected from the crime scene for the identification of the criminal. With the increase in crimes new techniques for investigation like Biometrics based on artificial intelligence are being developed.

Biometrics consists of two words ‘Bio’ meaning ‘life’ and ‘metric’ is the measurement or quantitative based recognition and identification. In simple words if you have to choose a T-shirt for your blue jeans what you

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would do? You would choose from a pile and match, much the same way in Biometrics where we compare given pattern from the existing ones in the data base. Basically, it is related to brain recognition with its stored observations and is faster with more accuracy.

4.1 BIOMETRICS: DEFINITION AND SYSTEM

Biometrics, in simple terms is identification of humans by their traits related to different aspects of physiology, chemistry or behaviour. Biometrics is defined as the applicable means of identifying and authenticating individuals in a reliable and fast way through the use of unique biological characteristics and technologies that lead to the acceptable way of solving the crime. Biometrics includes two types of characteristics:

- ❖ **Physiological:** face, fingerprint, hand geometry and iris recognition, and
- ❖ **Behavioural:** gait, odour, signature and voice.

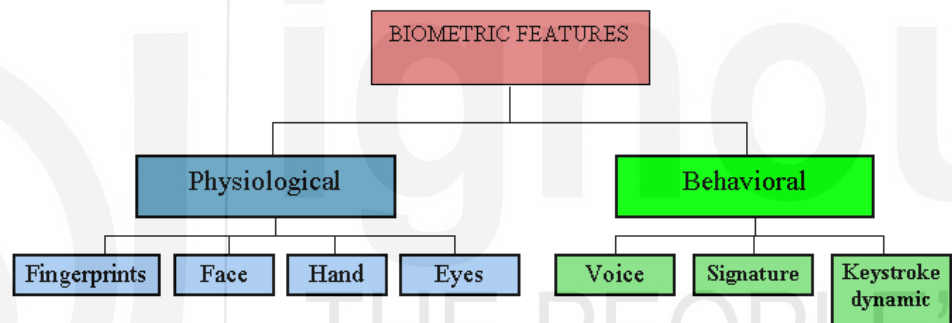


Figure1: Biometric Features

Source: http://www.biometrika.it/eng/wp_biointro.html

Biometric system use the following human characteristics as identifiers:

Common:

- 📱 Fingerprint Recognition
- 📱 Face Recognition
- 📱 Speaker Recognition
- 📱 Iris Recognition
- 📱 Hand Geometry
- 📱 Signature Verification

Others:

- 📱 DNA
- 📱 Retina Recognition
- 📱 Thermo Grams
- 📱 Ear Recognition

-  Skin Reflection
-  Lip Motion
-  Body Odour
-  Brain Wave Pattern
-  Footprint and Foot Dynamics

Biometric system consists of device which works as a system shown in figure 2.

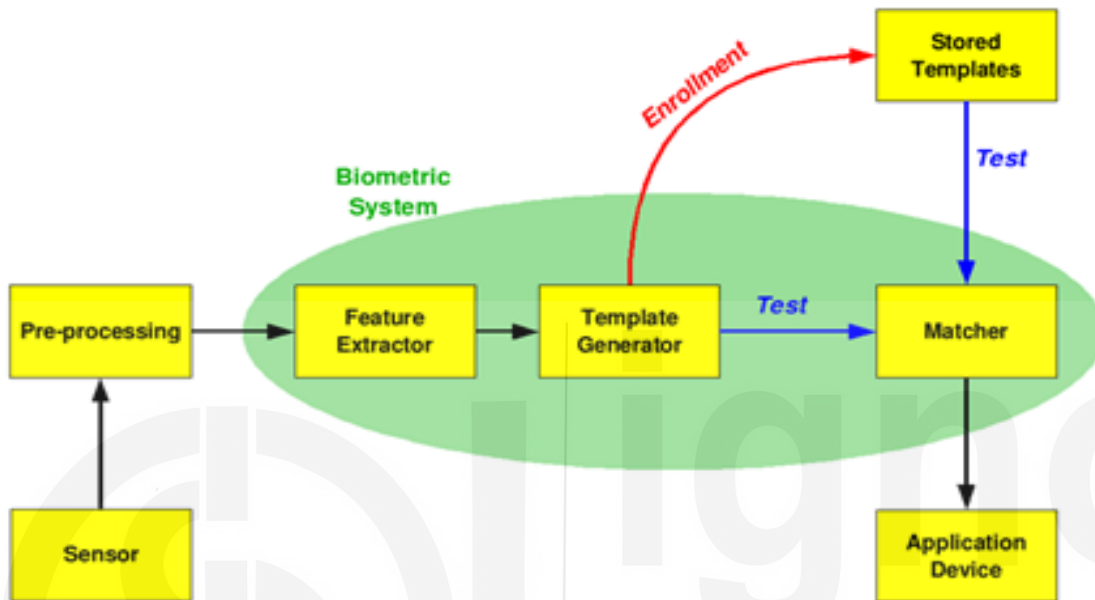


Figure 2: Biometric System

Source: <https://en.wikipedia.org/wiki/Biometrics>

From diagram you can see that a sensor recognises the information, then during pre-processing, it extracts information with template for matcher and identifies the information.

4.1.1 Advantages

Biometrics with any device, either system or characteristics are used with a common aim to collect the information and compare for identification. The advantages are:

- ❖ Universally found in all individuals and unique to differentiate one individual from another,
- ❖ Permanent, allowing for change over time
- ❖ Recordable (with or without consent)
- ❖ Measurable for future comparison
- ❖ Forgery-proof (a face, a fingerprint)
- ❖ Accuracy, improved security systems and quick authentication

4.1.2 Disadvantages

Biometric system has advantages nevertheless also has disadvantages like any other system. Some of the disadvantages are:

- Physical disabilities caused by injury, etc. leads to change the system
- Expensive
- Physical traits can't be altered
- Scanning is not reliable e.g. in retina scanning
- Software issues like power issues
- Remote access is lacking and security breaches

4.1.3 Applications of Biometrics

You would like to ask the question that when Biometrics has disadvantages then why it is developing. Biometrics has lot of commercial and daily life applications which creates innovative new systems which are precise and less erroneous. It is used in the:

- ✚ Identification and Authentication
- ✚ Legal applications: Justice and law enforcement
- ✚ Government applications
 - Border control and airport
 - Healthcare and Medical
- ✚ Commercial applications
 - Security
 - Attendance and Time Management
- ✚ Finance applications of biometric technology, financial identification, verification, and authentication in commerce to make banking, purchasing, and account management safer and more convenient and responsible.
- ✚ Mobile
- ✚ Eye movements tracking applications
- ✚ Automotive industry
- ✚ Screen navigation
- ✚ Aviation
- ✚ Detection and recognition of dynamic shapes
- ✚ Forensic Science Applications
- ✚ Physical and logical access (owner/user/employee/contractor/partner identification) identification
- ✚ Commercial applications (consumer/customer ion)

- 1) Define biometrics and state its importance in forensic science.

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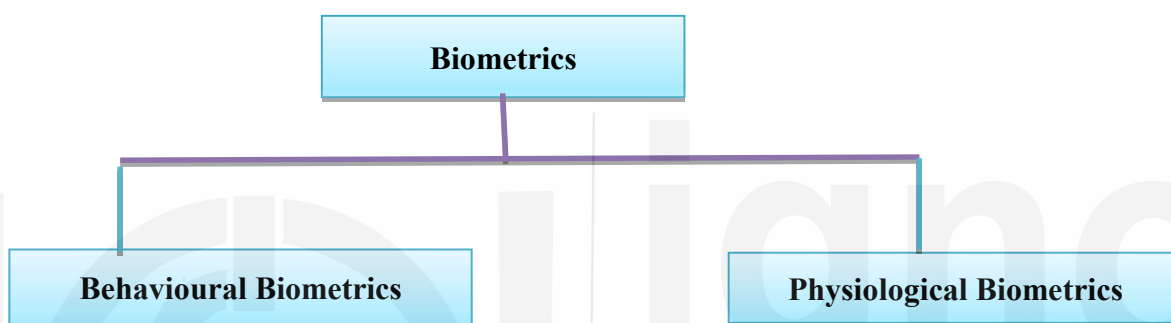
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4.2 TYPES OF BIOMETRICS



Behavioural Biometrics is the scientific study of body function and is divided as:

- ✚ **Signature Recognition:** The biometric signature recognition system verifies an authorised user for authentication and authorisation.
- ✚ **Voice Recognition:** Voice is a behavioural as well as a physiological trait that depends on the physical structure of the throat and mouth and helps to distinguish the speaker as per the speech.
- ✚ **Keystroke:** Identification on the basis of striking the keyboard and its rhythm.

Physiological biometrics is based on physical characteristics like, ears, eyes, iris, fingerprints, and is divided as:

- ✚ **Ear authentication:** It is based on sound waves to determine the way of the ear canal of an individual.
- ✚ **Eye vein recognition:** Biometric identification method by employing pattern formed by veins on the sclera (the white outer coating of the eye).
- ✚ **Facial recognition:** 3D structure of face recognition.
- ✚ **Finger vein recognition:** Blood vessels pattern recognition under the skin of fingers.
- ✚ **Fingerprint recognition:** Fingerprint recognition is comparing the finger prints with available data as no two persons can have similar fingerprints.

- ✚ **DNA matching:** The identification based on the analysis of segments from DNA.
- ✚ **Footprint and foot dynamics:** Footprint is a unique physiological character with distinctive properties useful in personal identification
- ✚ **Gait recognition:** A gait is defined as the cyclic and coordinated combination of movements for human locomotion and a person can be identified by observing his/her gait as a distinguishable feature.

4.2.1 Finger Print Biometrics

4.2.1.1 Introduction

Fingerprint is defined as featured pattern of one finger of ridges and valleys on the surface of a fingertip. The uniqueness in the fingerprint of every individual is based on local ridge characteristics not evenly distributed.



Figure 3: Finger Print

Source: <https://www.dreamstime.com/stock-images-fingerprint-image8262454>

The differentiation of the fingerprints is minutiae i.e. prominent local ridge characteristics or abnormal points on ridges as mentioned below:

- 1) Ridge ending: the point where a ridge ends abruptly.
- 2) Ridge bifurcation: the point where a ridge forks or diverges into branch ridges.

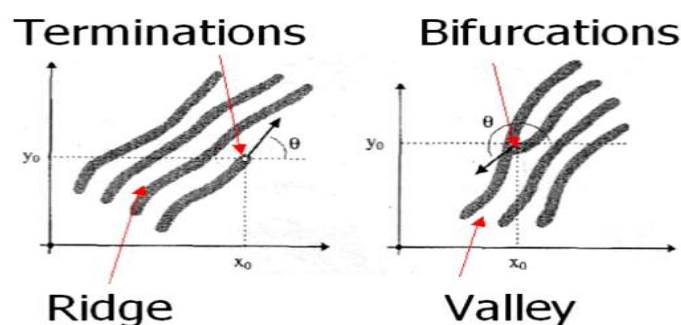


Figure 4: Ridge and Valley

Source: <https://www.semanticscholar.org/paper/Fast-Enhancement-Algorithm-with-Improved-Clarity>



Individual characteristic (Minutiae)

ridge ending	1		bridge	7	
bifurcation	2		double bifurcation	8	
dot	3		trifurcation	9	
island (short ridge)	4		opposed bifurcations	10	
lake (enclosure)	5		ridge crossing	11	
hook (spur)	6		opposed bifurcation/ridge ending	12	

Figure 5: Three Levels of Finger Print Classification

Source: https://www.researchgate.net/publication/281372646_A_Minutiae_Count_Based_Method_for_Fake_Fingerprint_Detection/figures?lo=1

The uniqueness of fingerprints is due to furrows and friction ridges present on fingers or thumb. On the basis of ridge pattern finger prints are classified into three types:

Whorl: The spiral patterns giving an appearance of whirlpools is called whorl and has four types- double loop i.e. S-shaped pattern, concentric plain circles, loop along with whorl i.e. central pocket loop and irregular shaped whorl.

Loop: The patterns that recurve back to form radial or ulnar loops depending on the pointing direction i.e. towards ulna or radius.

Arch: A wavy pattern i.e. tented (arch rise at sharp point) or plain arches.

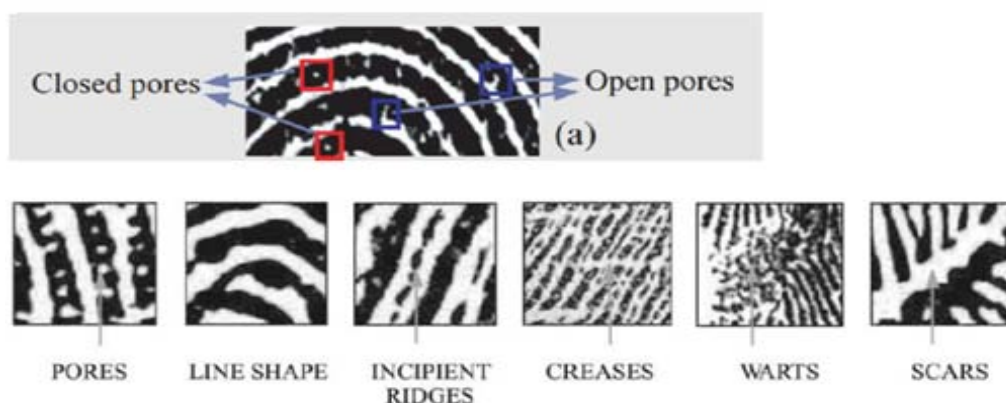


Figure 6: Pores and Ridges

Source: <https://www.semanticscholar.org/paper/Pores-and-Ridges%3A-Fingerprint-Matching>

4.2.1.2 Finger Print Biometrics Definition

Finger Print Biometrics is defined as the method of identifying finger print by employing the software on the basis of pattern of finger and location of the minutiae.

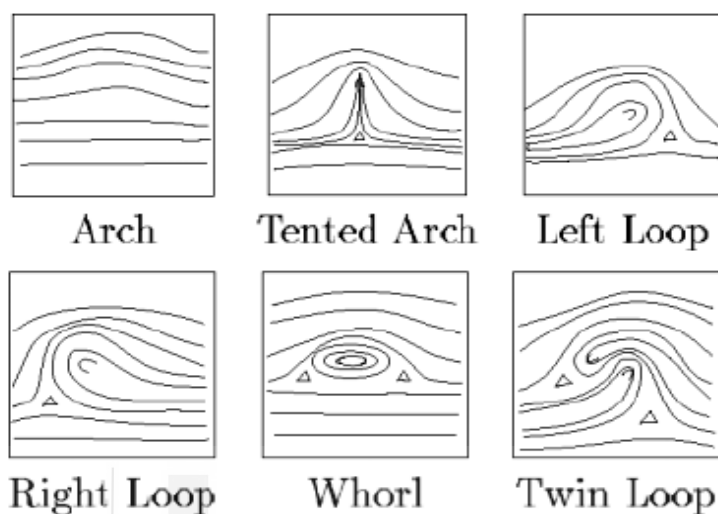


Figure 7: Six Patterns of Fingerprint

Source: <https://www.popsci.com/science/fingerprint-patterns-genetics>

Structure of Minutiae:

Crossover	two ridges cross each other
Core	center
Bifurcation	ridge separates
Ridge ending	end point
Island	small ridge between two spaces
Delta	space between ridges
Pore	human pore

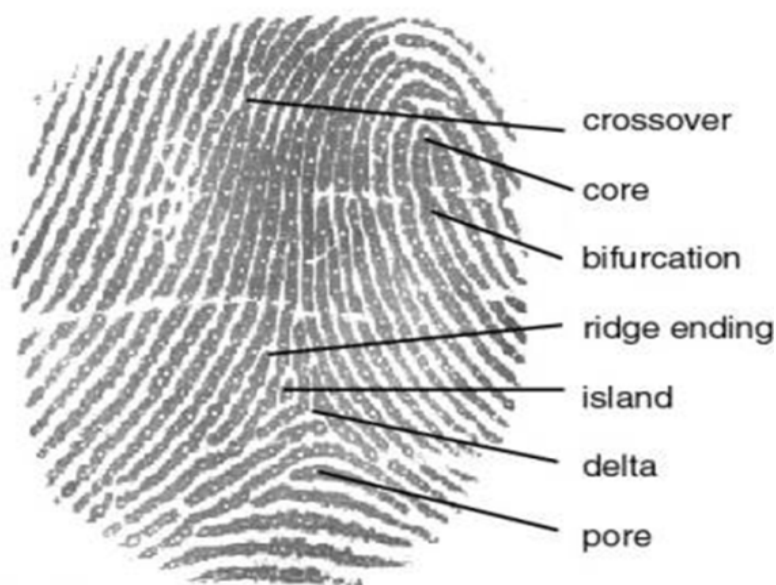


Figure 8: Structure of Minutiae

Source: <https://www.researchgate.net/>

4.2.1.3 Fingerprint Recognition (Physiological Trait)

The process of comparison between the known fingerprint against another in order to determine that it is of the same finger or palm is known as fingerprint recognition and consists of two parts.

Fingerprint Verification: to verify the authenticity of one person by his/her fingerprint.

Fingerprint Identification: to specify one person's identity by his/her fingerprint(s).

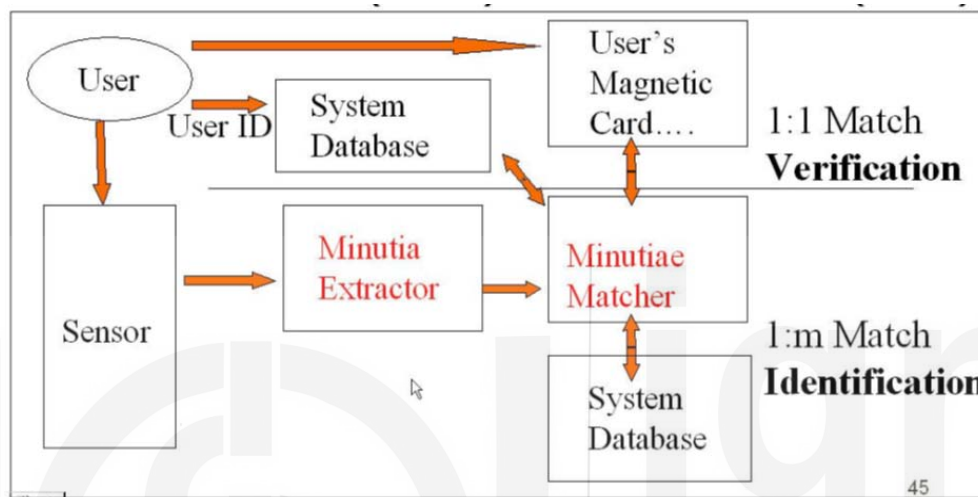


Figure 9: Fingerprint Recognition

Source: <https://electricalfundablog.com/fingerprint-recognition-system-architecture-works>

Three Stages of Fingerprint Recognition System

- 1) **Fingerprint Acquiring Device:** Optical or semiconductor sensors are employed for acquisition with high efficiency and acceptable accuracy while exceptions are dry or dirty fingers.
- 2) **Minutia extractions:**

Uses the ridge endings and bifurcations on a person's finger to plot points are known as Minutiae. The variations in the number and location of minutiae occurs from person to person, finger to finger in any particular individual



Figure 10: Extraction of Minutiae

Source: https://biometrics.mainguet.org/types/fingerprint/fingerprint_algo.htm

This level has three stages pre-processing, minutiae extraction and post processing stage.

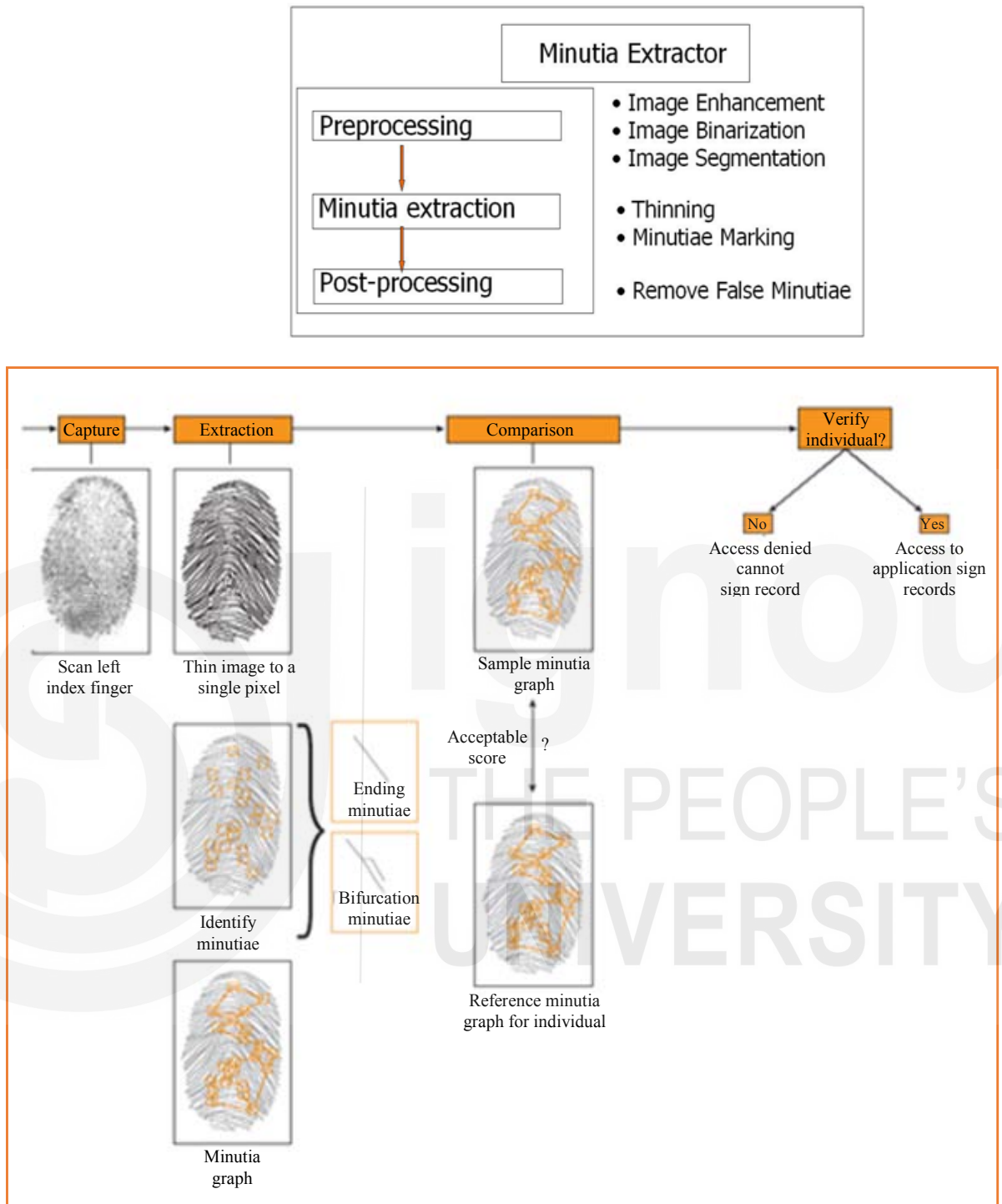


Figure 11: Finger Print Scanning

Source: <https://slideplayer.com/slide/13213653/>

3) Minutia Matching

Automatic finger print verification is achieved with minutia matching (point pattern matching) instead of a pixel-wise matching or a ridge pattern matching of fingerprint images. First of all, minutia matcher selects two minutiae as reference minutia pair after that matching of the ridges is done and finally on the basis of the ridge matching the image are aligned and matching of all minutiae is overseen.

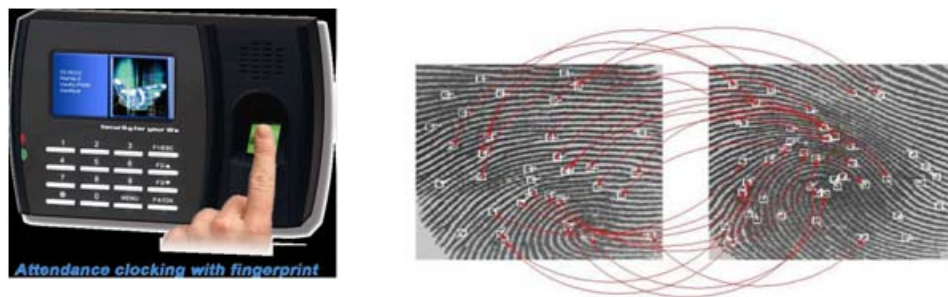


Figure 12: Finger Print Attendance and Minutiae Matching

Source: <https://slideplayer.com/slide/13213653/>

4.2.1.4 Advantages and Disadvantages of Fingerprint Biometrics

Advantages

- ❖ Fingerprint pattern remain stable throughout the lifetime, unique in nature.
- ❖ It is easily analysed and can be compared.
- ❖ Inexpensive device and oldest form of biometrics.

Disadvantages

- Wet or moist fingers, cut fingers, or dirt or grease can sometimes affect the authentication process.
- It is not right tool for those persons who work in chemical labs.

Applications of Finger Printing

- ✚ Banking Security - ATM security, card transaction
- ✚ Physical Access Control (e.g. Airport)
- ✚ Information System Security
- ✚ National ID Systems, Passport control (INSPASS), Voting
- ✚ Prisoner, prison visitors, inmate control
- ✚ Identification of Criminals and missing children
- ✚ Secure E-Commerce (Still under research)

A person's biometric identity is on the basis of one-to-one match or on other security measures like password, PIN etc.



Figure 13: Fingerprint Biometrics

Source: <https://slideplayer.com/slide/13213653/>

Check Your Progress 2

- 2) Explain fingerprint recognition with applications and advantages.

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- 3) Draw different patterns of fingerprint.

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4.2.2 Iris Patterns Biometrics

Eyes are important part of the individual and iris or the coloured part of the eye which has thick and thread-like muscles for Biometric recognition as shown in the figure 14.

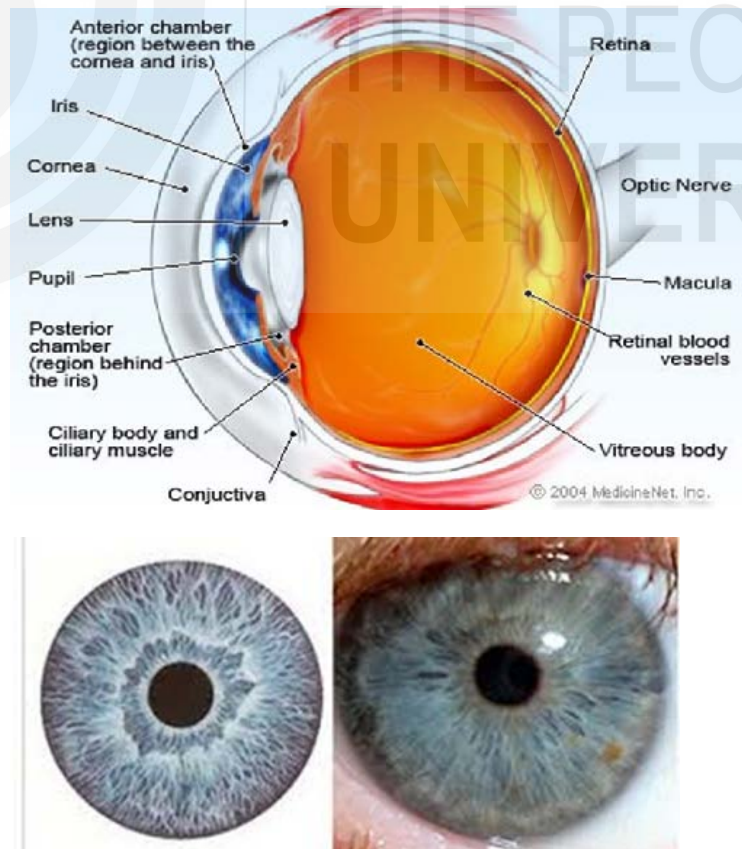


Figure 14: Iris Structure

Source: [https://en.wikipedia.org/wiki/Iris_\(anatomy\)](https://en.wikipedia.org/wiki/Iris_(anatomy))

Iris is made up of pigmented elastic tissue with an adjustable circular opening in the centre. The muscular pattern and colour is unique in every individual even the right and left eye is different thus it becomes a reliable biometric recognition character. Therefore you have to remember that the colour and muscular pattern is a feature for Iris Recognition along with distant technology applicable.

Iris recognition is an automated method of biometric identification by employing mathematical pattern recognition techniques on the images of the iris of an individual's eye or unique complex random patterns.

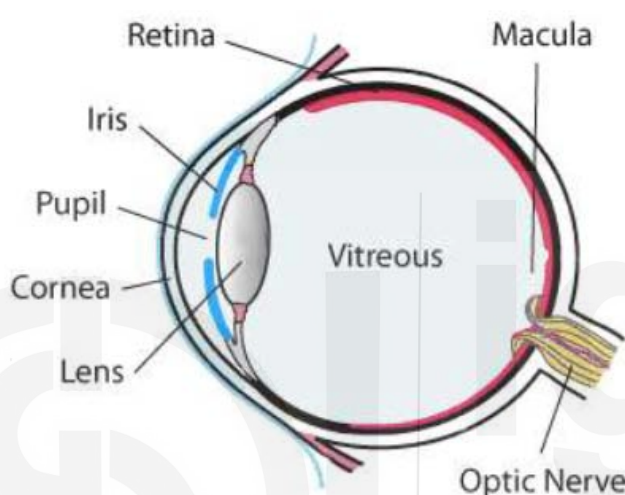


Figure 15: Eye Structure

Source: <https://www.merckmanuals.com/home/eye-disorders/biology-of-the-eyes/structure-and-function-of-the-eyes>

4.2.2.2 Iris as Biometric

Further you have to understand that the function of muscles is to provide shape to pupil and light that enters the eye. The uniqueness of the folds of muscles results into biometric authentication to confirm the identity with incredible accuracy. Other essentials you have to note is liveness detection like blink for scan enhance its accuracy and security. The technology has advantage of its applicability from a distance also. Note that the iris pattern is not subject to alteration with age, eye redness or alcohol consumption. Iris recognition can be used as Biometric because iris is the coloured portion of the eye surrounded by the pupil and its pattern is the result of meshwork of muscle ligament while colour and contrast are due to pigmentation.

4.2.2.3 Iris Recognition (Physiological Trait)

The recognition of Iris is carried out on the basis of the following traits:

- ✚ Iris scanning measures the iris pattern in the coloured part of the eye.
- ✚ Analysis of the iris of the eye, which is the coloured ring of tissue that surrounds the pupil of the eye.

- Based on visible features.
- Widely regarded as the most safe, accurate biometrics technology
- High speed, high accuracy.

HOW IRIS SCANNERS RECORD IDENTITIES

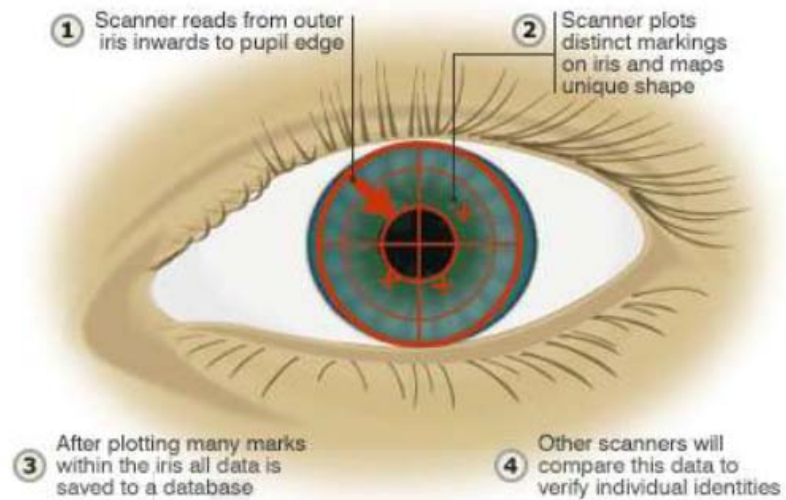


Figure 16: Eye

Source: <https://www.merckmanuals.com/home/eye-disorders/biology-of-the-eyes/structure-and-function-of-the-eyes>

4.2.2.4 Iris Recognition Systems

Biometric Iris recognition is based upon the working of camera with the mathematical analysis of the random patterns. Imagine Iris as your phone camera where camera and light combine to take a photo. In the same way iris camera carries out a person's identity by combining computer vision, pattern recognition, statistical inference and optics.

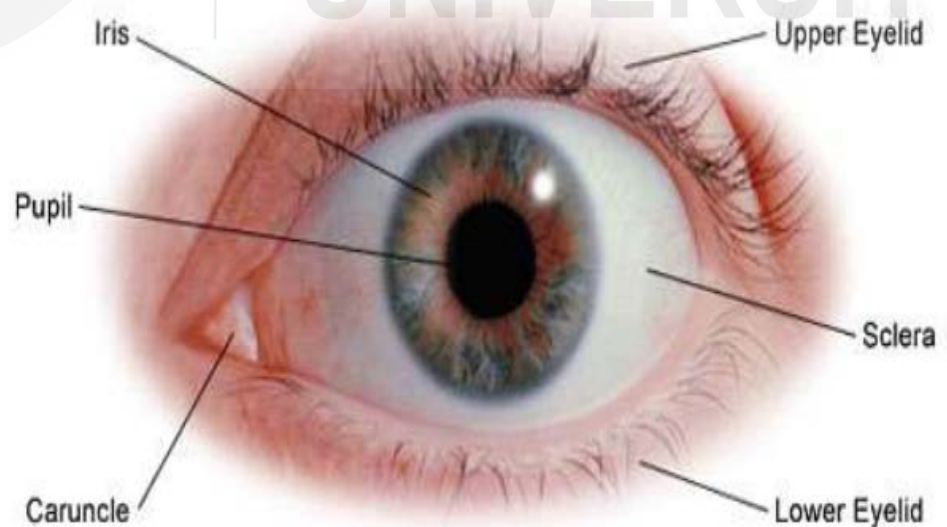


Figure 17: Eye Structure from Front

Source: <https://www.dreamstime.com/structure-eye-anatomical-external-structure-iris-cornea-eye-pupil-structure-eye-anatomical-image168611026>

This is how it happens:

- 1) A person stands in front of the iris identification system with a distance of one or three feet so that the wide angle camera calculates the position of their eye.
- 2) The function of the second camera is to zoom on the eye and takes a black and white image.
- 3) The focus of the iris overlays a circular grid on the image of the iris thus identifying the light and dark areas, like an “eye print”.
- 4) Fake eye prevention or fool system is done by the devices which vary the light shine into the eye and watch for pupil dilation.

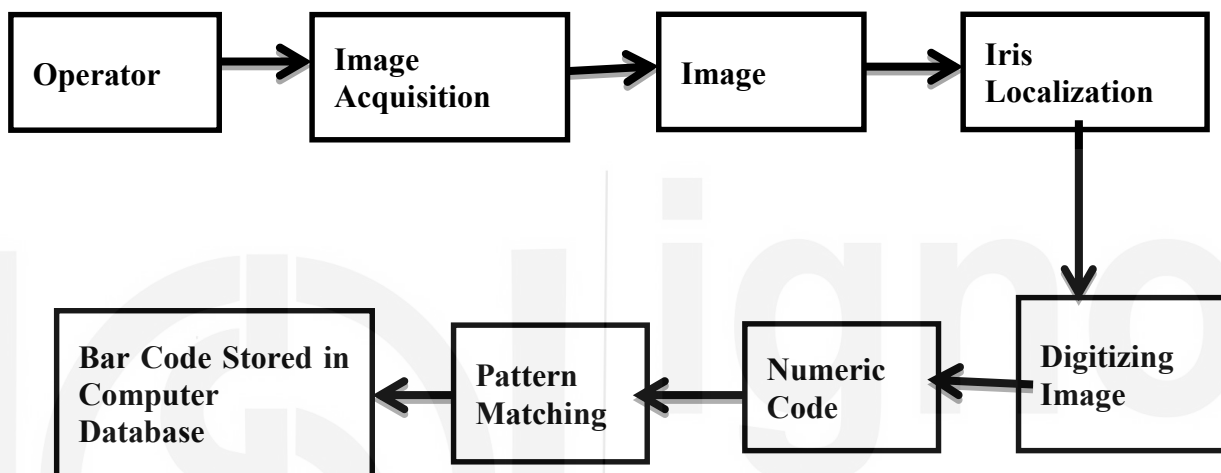


Figure 18: Block Diagram of Iris Recognition

Source: https://www.researchgate.net/publication/224597357_Iris_Segmentation_Using_Geodesic_Active_Contours/

4.2.2.5 Iris Recognition Function Steps

Four functional steps for Iris recognition are

- 1) Image Acquisition
- 2) Pre-processing
- 3) Image Analysis
- 4) Image Recognition

1) Image Acquisition

This is an important step as image acquired would lead to recognition and the characteristics required are:

- Good contrast,
- high quality image and
- high illumination.

Camera should be able to capture volume width on the order of 20–30 cm and more than 200 pixels across the iris for good quality as 100–150 pixels

results for marginal quality. Daughman's (1993) Algorithm described the methodology for image analysis i.e. Iris Segmentation as described: Iris is mapped into a rectangle in normalised polar coordinate system and segmentation normalises for scale change and pupil dilation.

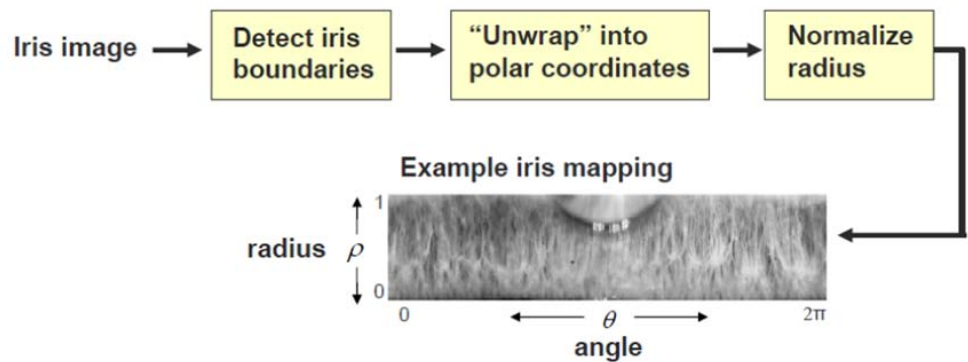
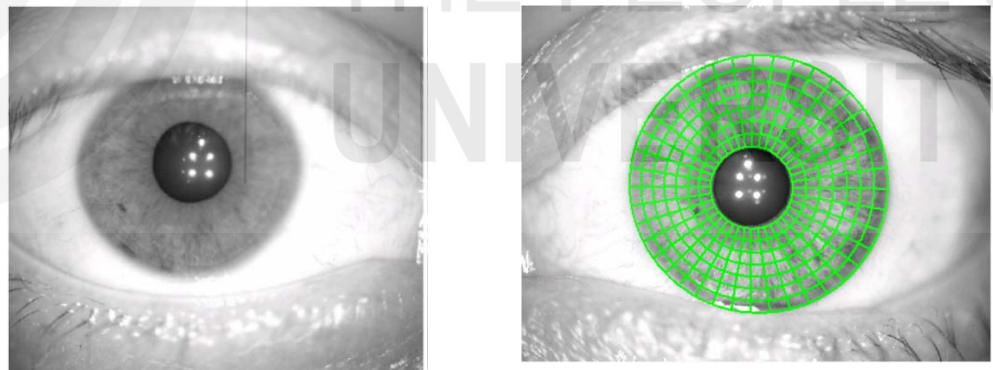


Figure 19: Mapping of Iris

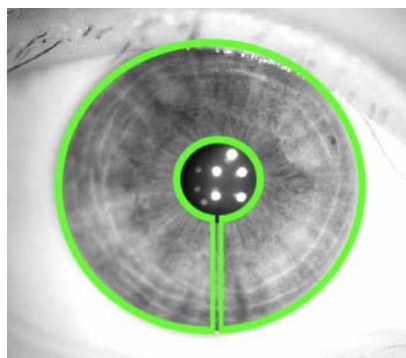
Source: https://www.researchgate.net/publication/224597357_Iris_Segmentation_Using_Geodesic_Active_Contours

Segmentation: Boundary Detection

- ✚ Segmentation is simplified by modelling the inner and outer iris boundaries as non-concentric circles.
- ✚ For each boundary, we must find 3 parameters: x and y of centre, and radius r.
- ✚ Search Criteria is based on intensities along with an expanding circular contour that become suddenly brighter (from red circles to green circles).

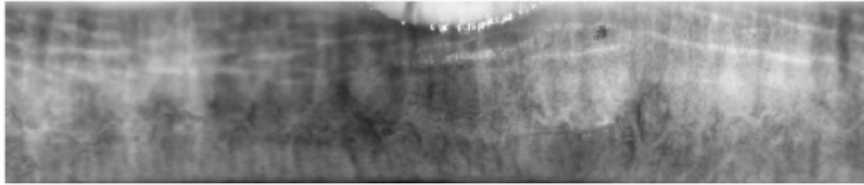


First Step: Iris Segmentation





Outer boundary (with sclera)



Inner boundary (with pupil)

Figure 20: Boundary Detection Example

Source: <https://casvisualfoundation.wordpress.com/2015/03/24/tuesday-excursion-research-and-experiment-4/>

2) Pre-processing

Next step is to classify whether the image acquired is useful or not which means the irrelevant parts e.g. eyelids or pupil and the pre-processing would remove the effect of the spots /holes lying on the pupillary area. This process also transforms the true colour into intensity image. The two steps of pre-processing are:

a) Iris Localisation

The outer and inner boundary of the Iris is considered as two circles that are not co-centric thus inner boundary between the pupil and the iris is detected. To detect the outer boundary due to its low contrast between the two sides is found by maximizing changes of the perimeter- normalised along the circle.

b) Edge Detection

The usage of Edge Detection is to evaluate complex object boundaries by marking potential edge point corresponding to places in an image where rapid change in brightness occurs and edge is defined by the discontinuity in grey values thus an edge separates two distinct objects.

3) Image Analyses

Image analysis visualises the features of Iris and digitised into a 512 byte (4096 bits) Iris Code record. Half Iris Code describes the features and another half controls the comparison process.

4) Image Recognition

The database is built by storing the different Iris Code. You can apprehend that the recognition is same as going to the market to buy a T-shirt of

particular colour by searching and comparing. Same way an iris is presented at a recognition point and process is repeated and compares it to every file in the Database Recording of Identities thus the Iris Code record is not stored.

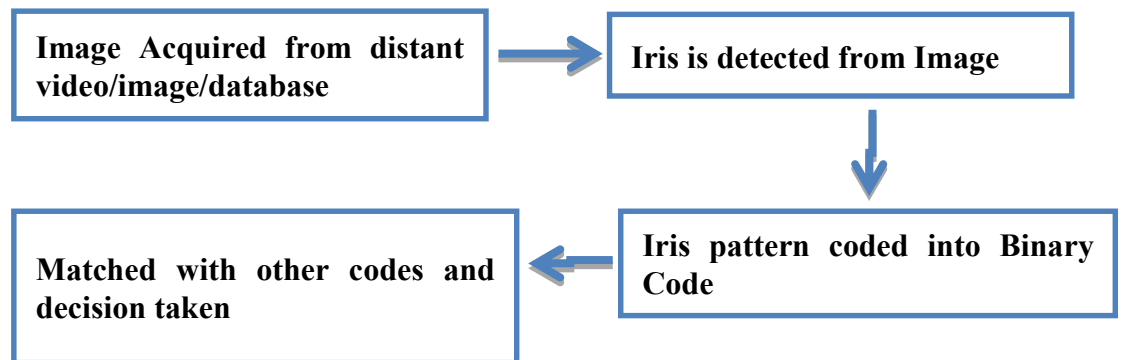


Figure 21: The Process Overview

4.2.2.6 Advantages of Iris Technology

Advantages of the technology are mentioned below:

- ❖ Accuracy: Uniqueness of iris patterns hence improved accuracy.
- ❖ Highly protected: It is the internal organ of the eye.
- ❖ Stability: Persistence of iris patterns.
- ❖ Non-invasive: Relatively easy to be acquired.
- ❖ Unique: The probability of two rises producing the same code is nearly impossible.
- ❖ Flexible: Iris iris recognition technology easily integrates into existing security systems or operates as a standalone.
- ❖ Stability: Patterns apparently stable throughout life.
- ❖ Reliable: A distinctive iris pattern is not susceptible to theft, loss or compromise.

4.2.2.7 Disadvantages of Iris Technology

Iris Biometrics disadvantages are listed below:

- ✚ Small target (1 cm) to acquire from a distance (1m).
- ✚ Alcohol consumption causes deformation in Iris pattern.
- ✚ Illumination should not be visible or bright.
- ✚ Obscured by eyelashes, lenses, reflections.
- ✚ Difficult to capture an image of handicap people sitting on wheel chair as cameras are usually attached on the wall therefore can capture an image up to a certain height.
- ✚ Costlier iris recognition systems.
- ✚ Authentication is affected with glasses or direct sunlight.

4.2.2.8 Applications of Iris Recognition System

Iris recognition is playing an important role:

- ✚ in providing unique identification to the individuals for the bank accounts aadhar card etc.
- ✚ as living password for computer login.
- ✚ as living passport for national borders security.
- ✚ driving licenses and personal certificates.
- ✚ internet security, control of access to privileged information.
- ✚ premises access control (home, office, laboratory).
- ✚ anti-terrorism (e.g. security screening at airports).
- ✚ financial transactions (electronic commerce and banking).
- ✚ iris password for the bank cash machine accounts.
- ✚ credit-card authentication.
- ✚ automobile ignition and unlocking; anti-theft devices.

Real Life Applications

- ❖ Iris scanning forms the basis for Aadhaar Card.
- ❖ Border Patrolling in UAE.
- ❖ Iris scanning permits the passport in countries US, Canada.
- ❖ Iris scanners are employed to control access of Datacentres of Google.

Check Your Progress 3

- 4) Explain in detail about the image acquisition in iris scan technology.

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- 5) Draw the structure of Iris.

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4.2.3 Retinal Pattern Biometrics

Dr. Carleton Simon and Dr Isodore Goldstein in 1930-1935 conducted first scientific study to describe how every individual possesses a unique Retinal structure. Because of the differing distribution of blood vessels, they suggested the use of the Retina to confirm the identity of an individual. Dr. Paul Tower (1950) reported that even identical twins have a very distinct and unique set of Retinas and has DNA Strand leading to very stable Biometric modality and that hardly changes.

Human retina is a thin tissue composed of neural cells located in the posterior portion of the eye. Retina is a complex structure with a network of blood capillaries and vessels resulting into the unique feature for every person as identical twins do not have same features. Retinal patterns are altered due to medical morbidities otherwise it is same from birth to death. The unique Retinal patterns of the individual is used as biometric identifier and the blood vessels within the retina absorb light more readily than the surrounding tissue which forms the typical smallest template for the Biometric Technology. The principle for the Retinal scan is carried out by casting an unperceived beam of low-energy infrared light (IR) into a person's eye and beam traces a standardised path on the retina as retinal blood vessels are more absorbent than the rest of the parts of the eye, then this path is converted to computer mode.

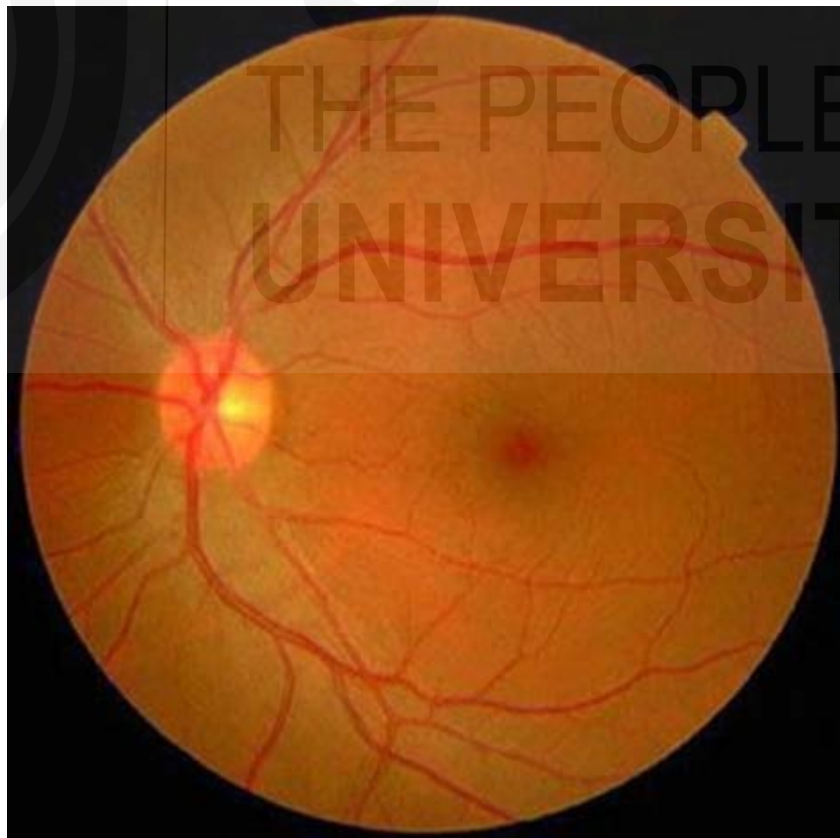


Figure 22: Retina Structure

Source: <https://jackwestin.com/resources/mcat-content/vision/structure-and-function-of-the-eye>

4.2.3.1 Process of Retinal Recognition

In comparison to Iris recognition where images are collected from a far distance but Retinal recognition requires end user must be in close proximity to retinal scanning device. The process has following steps:

1) Image/Signal Acquisition and Processing:

The individual eye resides on tiny receptacle to receive infrared light beam for the illumination of retina at 360°. The duration of the process is few minutes with precaution that the individual must remain still till one rotation is completed. Five images are taken to create the composite image from which the unique features are extracted.

2) Enrolments and Verification Template Creation:

Retinal patterns are not genetically controlled thus 400 unique data points are captured from the Retina, unlike the fingerprint, where only 30-40 unique data points are captured to create the Enrollment Template with a size of 96 bytes. The small size makes it possible to store number of templates in a single database and same process leads to creation of Verification Template. Due to the close proximity of numbers of factors influence the quality of Enrollment and Verification Templates as mentioned below:

- ✚ Individual cooperation is lacking
- ✚ The individual's eye and the lens distance is more.
- ✚ The size of the individual's pupil:



Figure 23: Retinal Recognition

Source: <https://platform.keesingtechnologies.com/retinal-recognition-pros-and-cons/>

4.2.3.2 Advantages and Disadvantages

The advantages of Retinal Recognition are:

- ❖ Due to stability and hardly ever changes over the lifetime an individual retina is a reliable Biometric technology.
- ❖ Small sized images give quick results for the identity of the individual.
- ❖ More number of unique data points and makes error less identity.
- ❖ Retina is present within the eye thus not affected by the external environment like fingerprints.

Disadvantages

- ✚ The individuals are not comfortable to place their eye on the receptacle with the direct infrared light.

- ✚ Retinal Recognition requires high level of cooperation and motivation from the end user to capture high quality, raw images leading to low verification with a metric of 85% as compared to 100%.
- ✚ The cooperation of the end user leads one to take numerous attempts and a long time till the final result is obtained therefore there are many false rejections.

4.2.3.3 Criteria for Effectiveness and Viability of Retinal Recognition

Overall, the effectiveness and viability of Retinal Recognition is examined against seven criteria as mentioned below:

- ✚ Universality
- ✚ Uniqueness
- ✚ Permanence
- ✚ Collectability
- ✚ Performance
- ✚ Acceptability
- ✚ Resistance to circumvention

The retinal template generated is typically one of the smallest of any biometric technology.

4.2.3.4 Similarities and Differences:

While both iris and retina scanning are ocular based biometric distinct technologies but there are evident similarities and differences as mentioned below:

Similarities:

- Low occurrence of false positives.
- No two individuals have the same Iris and Retina pattern thus reliable technique.
- Identity of the individual is authenticated very fast.
- The capillaries in the iris and retina decompose too rapidly to use an amputated eye to gain access.

Differences:

- Retinal scan measurement accuracy can be affected by disease; iris fine texture remains remarkably stable.
- Taking a photograph is similar to iris scan from distance while for the retinal scanning the individual has to be near like seeing a slide under microscope.

- Iris scanning is more widely accepted as a commercial modality than retinal scanning
- Retinal scanning is invasive but not iris.

The primary function of Iris is to allow the requisite amount of light required for proper vision. The retina is located at the back of the eye and links front of the optic nerve to the brain therefore the function of retina is to send visual information and cues to the brain for processing.

In other words, at Retina, image is formed that means it acts like a film of camera.

Check Your Progress 4

- 6) Write a short note on retinal scanning technology.

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4.2.4 Facial Recognition Biometrics

Another Biometric Technology includes the Face Recognition a technique that is capable of identifying or verifying a subject through an image. This method employs measures of face and head to identify the identity of a person through the facial biometric pattern. The first facial recognition was carried out in 1960 based on the features of eyes, nose, etc. The 21 subjective markers e.g. hair colour, lip thickness were employed by Goldstein and Harmon in 1970 to automate the recognition. The standard linear algebra techniques were applied in 1988 by Kirby and Sirovich for face recognition. The reasons that Face Biometric is beneficial than the Biometrics are:

- ✚ No physical interaction on behalf of the user is required.
- ✚ Accurate with high enrolment and verification rates.
- ✚ No experts as such needed for interpretation of the results.
- ✚ Only biometric that allows passive identification in a one to many environments.

4.2.4.1 Facial Recognition System

Two types of comparisons are done

Verification- It is comparison of the given individuals who are similar by the system and then result is yes or no decision.

Identification- The given individual is compared to all the entities present in the database thus deciphering a rank list of matches.

There are four stages in Identification technologies i.e.

- Capture:** During enrolment or verification or identification process, the system captures a physical or behavioural sample.

- ii) **Extraction:** The creation of template by extracting unique data.
- iii) **Comparison:** Compare the template with a new sample.
- iv) **Match/non-match:** Matching the features extracted from new sample and decide whether matched or not.

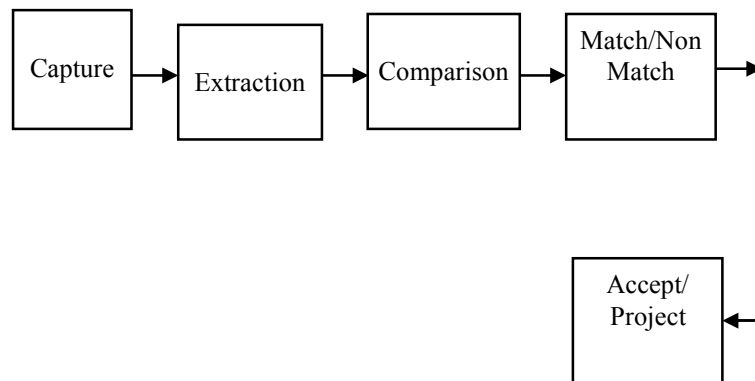


Figure 24: Four Stages of Identification

Source:https://www.researchgate.net/publication/328577882_Face_Recognition_Based_on_PCA_and_DCT_Combination_Technique/

4.2.4.2 Components of Face Recognition Systems

Enrolment Module: An automated mechanism that scans and captures a digital or an analog image of living personal characteristics

Database: Another entity which handles compression, processing, and storage of the captured data with stored data

Identification Module: The third interfaces with the application system

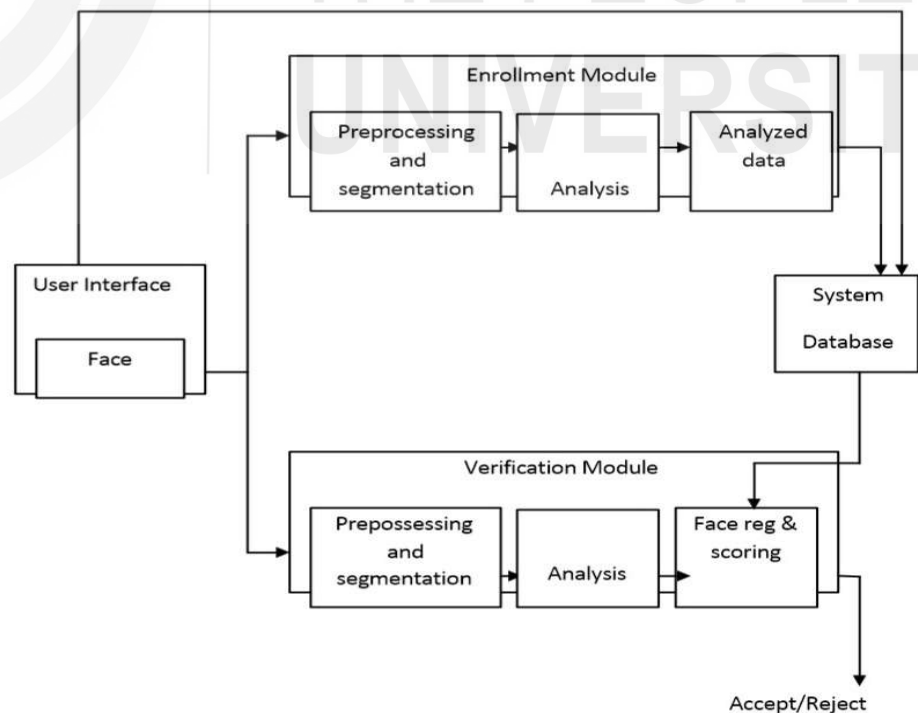


Figure 25: Components of Face Recognition

Source:https://www.researchgate.net/publication/328577882_Face_Recognition_Based_on_PCA_and_DCT_Combination_Technique/

4.2.4.3 Implementation of Face Recognition System

There are four stages in implementation process as mentioned below:

- i) Image Acquisition
- ii) Image Processing
- iii) Distinctive Characteristic Location
- iv) Template Creation And Matching

i) Image Acquisition

The static camera or video system acquires image of quality and resolution, an essential component for verification, identification and authentication.

ii) Image Processing

The image processing steps:

- Detect the face or faces at crime scene
- Normalise and localise the face to bring the dimensions in live facial sample
- Crop the image to get ovoid facial image
- Coloured images conversion to black and white
- Initial comparisons on grey scale basis
- Align the facial sample with the templates

Distinctive Characteristic Location

The features used for matching are upper ridges of the eye sockets, areas around the cheekbones, sides of the mouth, nose shape, and the position of major features relative to each other and these features hardly change.

Template Creation and Matching

Processed facial images help to create enrolment template of varying size from 100 bytes to 3K and larger templates are related to behavioural biometrics. Matching is done 1:1 verification within facial scan system and it is not as effective as other scans and many are returned.

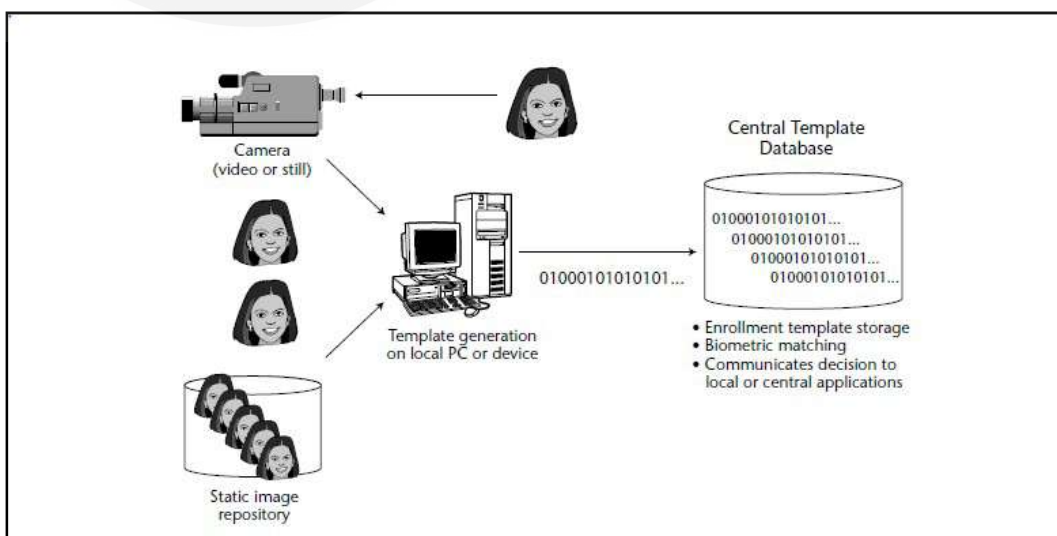


Figure 26: Template Creation

Source: https://www.researchgate.net/publication/328577882_Face_Recognition_Based_on_PCA_and_DCT_Combination_Technique

4.2.4.4 Face Recognition Systems Working

Everyone has distinguishable points on the face i.e. valleys and peaks which are defined as nodal points in software. There are 80 nodal points on human face that are measured by software i.e.

-  distance between the eyes
-  width of the nose
-  depth of the eye socket
-  cheekbones
-  jaw line
-  chin, etc.

For the general representation of face 70 nodal points are used to construct the face bunch graph and an image template is matched with this face graph in order to get the same points. The measured nodal points are given numerical code and strings of numbers represent a face in the database called face print.



Figure 27: Face Bunch Graph

Source: <https://www.semanticscholar.org/paper/Face-Recognition-for-Smart-Environments-Pentland>

4.2.4.5 Strengths and Weaknesses of Face Biometrics

The advantages of face biometrics are as:

- ❖ convenient and has social acceptability as only picture is required.
- ❖ person is not required only image can do.
- ❖ most inexpensive biometric and has leverage for existing image.
- ❖ searching can be done on basis of static image like license and can be performed without user cooperation.

This system has one weakness:

- cannot differentiate between identical twins.

4.2.4.6 Applications

Face recognition biometrics has replaced PIN naturally and other applications as listed below:

Government Use:

- Law Enforcement
- Security/Counter terrorism

- Immigration: Rapid progression through customs
- Voter verification

Commercial Use:

- Residential Security: Alert homeowners of approaching personnel
- Banking using ATM: Customer's face identified quickly with the software
- Physical access
- Face Recognition technology has now progressed and become cost effective, reliable and highly accurate thus being applied at many places.

4.2.5 DNA Biometrics

Behavioural or physiological traits in Biometric systems are pertaining to individual recognition and out of which DNA profiling and fingerprinting is the most reliable accurate method as it is intrinsically digital, and not altered during lifetime or death. DNA profiling is known as ultimate identifier.

DNA Biometrics is defined as a specific pattern obtained from a person or sample from the crime scene. DNA fingerprinting, is also known as DNA typing, DNA profiling, genetic fingerprinting, genotyping, or identity testing and it is the method of isolating and identifying variable elements within the base-pair sequence of DNA (deoxyribonucleic acid). British geneticist Alec Jeffreys (1984) developed the DNA Fingerprinting with a principle of isolating and identifying variable elements within the base-pair sequence of DNA which do not participate in functioning and highly variable called minisatellites. Jeffrey reported that unique pattern of minisatellites are present in every individual and only exception is identical twins from single zygote. At present DNA fingerprinting is making usage of microsatellites of short tandem repeats (STR) of 2-5 base pairs long.

Every individual has same chemical structure with a unique set of genes and 1% of DNA differs from person to person in sequence arrangement and it can be understood as STOP and POTS use same letters but meaning differ. DNA profiling is dependent on VNTR (Variable Number Tandem Repeats) which are of two types: minisatellites and microsatellites. DNA profiling uses repetitive sequences. VNTR loci are very similar between closely related individuals. The analysis of variable number of tandem repeats (VNTRs) is to detect the degree of relatedness to another sequence of oligonucleotides, making them ideal for DNA finger printing. There are two principal families of VNTRs:

a) Microsatellites

b) Minisatellites

Microsatellites are defined as repeating sequences of 2-6 base pairs of DNA which are also called Simple Sequence Repeats (SSRs) or Short Tandem Repeats (STRs). VNTR of 10-60 base pairs are named as minisatellites and their analysis has direct application genetics, forensics, biology and DNA fingerprinting.

4.2.5.1 Technique of DNA Fingerprinting

Step-1 Collection and Isolation of DNA

Reference Sample i.e. DNA sample of an individual of buccal swab or blood or biological samples collected from the crime scene. Significance of DNA analysis is that very small amount of sample is required in microliters. Alkali and salt treatment is done on sample to get purified DNA without proteins.

Step-2 Cutting, Size and Sorting

DNA is cut into small fragments by employing Restriction Enzymes which cut DNA at specific sites e.g. EcoR1 will cut DNA when the sequence 5'..GAATTC..3' occurs.



Figure 28: Each Enzyme cuts at Specific Site

Source: [sigmaaldrich.com/IN/en/product/sigma/r0260](https://www.sigmaaldrich.com/IN/en/product/sigma/r0260)

Cutting of DNA yields short and long fragments which are sorted according to size

Step-3 Separation of Fragments

Agarose gel electrophoresis is used to separate the DNA fragments on the basis of size. The negatively charged DNA moves towards positive end and shorter segments move faster than the longer thus DNA is sorted on the basis of size. The electrophoresis to sort more than one sample at one time as different samples can be loaded in the wells. Ethidium Bromide is added to get fluorescent bands which can be visualized under UV illuminator.

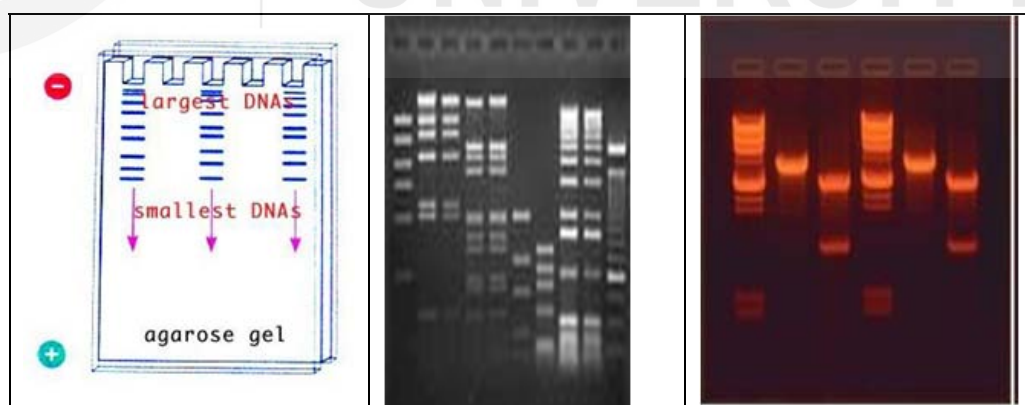


Figure 29: Agarose Gel Electrophoresis and Bands on the Gel

Source: https://www.researchgate.net/publication/280232114_Isolation_and_Characterization_of_Mercury_Resistant_Bacteria_from_Fly_Ash_Sample_of_Mejia_Thermal_Power_Plant_W_B_India_for_Application_in_Bioremediation_and_Phytoremediation/figures?lo=1

Step-4 Southern Blotting and Hybridisation

A Molecular technique used to transfer DNA from gel to nylon nitrocellulose membrane by applying blotting technique. Radioactive or coloured probes are added to nylon sheet which sticks to only one or two specific places thus probe is hybridised and pattern developed is DNAs fingerprint.

Step-5 DNA Fingerprint

The final DNA fingerprint is built by using several probes (5-10 or more) simultaneously and the bar codes are used like in shops. DNA Fingerprint is then analysed by comparing the victim and the suspect. The matching of the profile generates strong evidence that the suspect was at the crime scene but does not hint that he/she has committed the crime as it does not match, then eliminated from the enquiry.

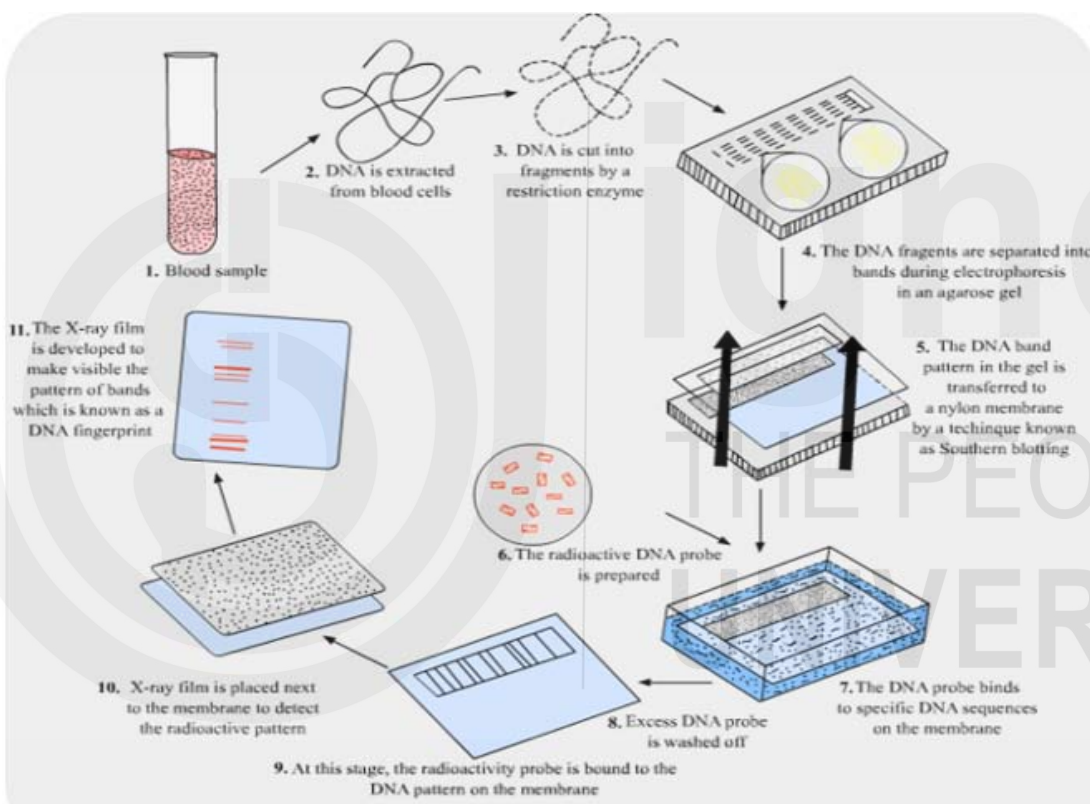


Figure 30: Steps in DNA Fingerprinting

Source: <https://allaboutdnafingerprinting.weebly.com/steps-of-dna-fingerprinting.html>

4.2.5.2 Applications of Fingerprinting

Diagnosis and Developing Cures for Inherited Disorders:

DNA fingerprinting is used to diagnose inherited disorders in both prenatal and new-born babies in hospitals around the world along with the probability of occurrence of the disease.

Biological Evidence to Identify Criminals:

DNA Fingerprinting is helpful in forensics as the biological specimens present at the crime are important source of DNA and proves to be valuable source of DNA of the criminal and innumerable cases are solved on this basis.

- 3) **Paternity Disputes:** DNA Fingerprint is utilised to solve paternity dispute cases in court as fingerprints unprecedented with accuracy and reliability.
- 4) **Personal Identification:** As same DNA is present in all the body tissues and organs thus DNA profiling is the best tool for personal identification

4.2.5.3 Advantages and Disadvantages of Fingerprinting

Advantages of DNA matching

- ❖ Accurate method of identification and has become a reliable and matured method of identification.
- ❖ Used for medical purposes e.g. the success of bone marrow transplant is dependent on DNA matching.
- ❖ No system performance issues.

Disadvantages of DNA matching

- 📁 DNA profiling requires a physical sample for collecting biometric data and has to be stored with appropriate environmental conditions.
- 📁 DNA Samples can be prone to contamination.
- 📁 It is not fully automated yet and fast unlike other biometric methods even expensive.

Let us see the Comparison

DNA	Conventional Biometrics
Requires an actual physical sample	Uses an impression, image, or recording
Not done in real-time; not all stages of comparison are automated	Done in real-time; automated process
Does a comparison of actual samples	Uses templates or feature extraction

Check Your Progress 5

- 7) Draw a diagram of the procedure of DNA fingerprinting.

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8) How can DNA be a useful tool in the forensic applications?

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4.2.6 Handwriting and Voice Biometrics

Voice recognition is a crucial Biometric identifier as it recognises the speaker as well as the speech. The measurable, physical characteristic or personal behavioural traits are applied to verify and authenticate an individual. The two components, Speech recognition and Speaker verification are different as former identifies what you are saying and the later verifies who you are.

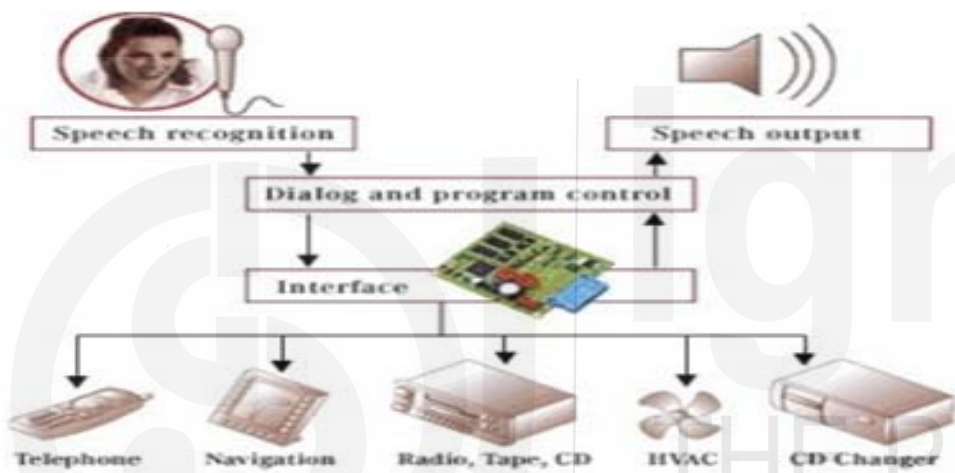


Figure 31: Speech Recognition

Source: https://www.researchgate.net/publication/334285447_Project_Report_On_AI_Speech_Recognition_System/

A speech recognition system assists the speaker in accomplishing what that person wants to do and to identify the speaker then the system has to authenticate the person and verify that he or she is in fact that person.

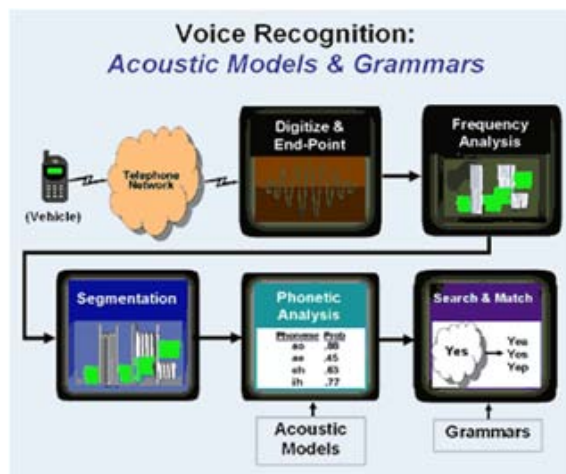


Figure 32: Voice Recognition Model

Source: <https://developer.nvidia.com/blog/how-to-build-domain-specific-automatic-speech-recognition-models-on-gpus/>

Biometric System is the combination of speech recognition and speaker verification systems

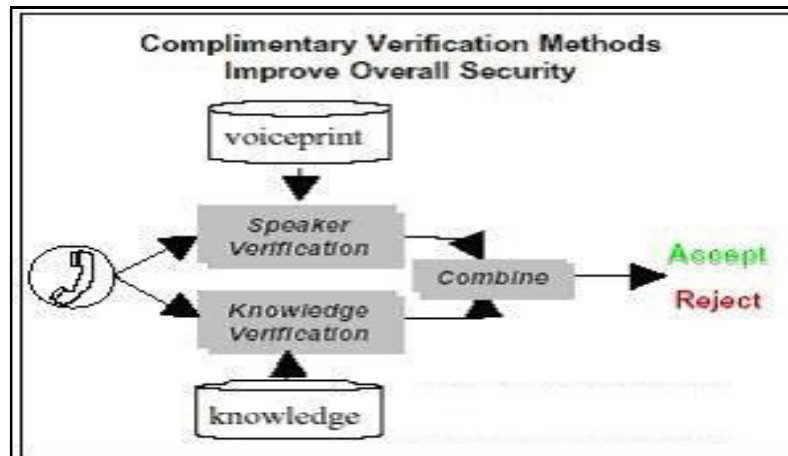


Figure 33: Biometric System

Source: <https://www.researchgate.net/>

Voices have unique features like frequency and intensity which is manipulated in the system to digitising a person's speech to produce a “voice print”. See figure 35.

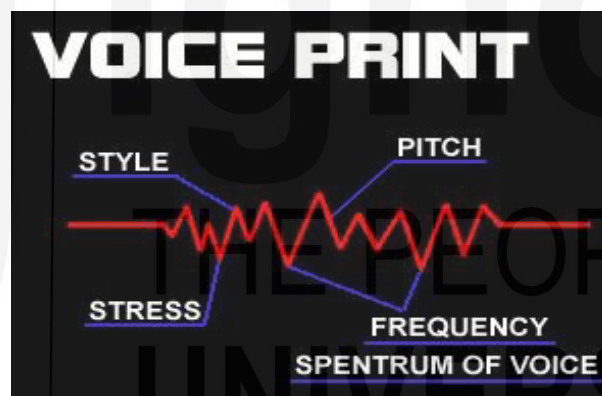


Figure 34: Voice Print

Source: <https://developer.nvidia.com/blog/how-to-build-domain-specific-automatic-speech-recognition-models-on-gpus/>

In the second step these smaller segments are “captured” in a digital form and plotted on a spectrum.

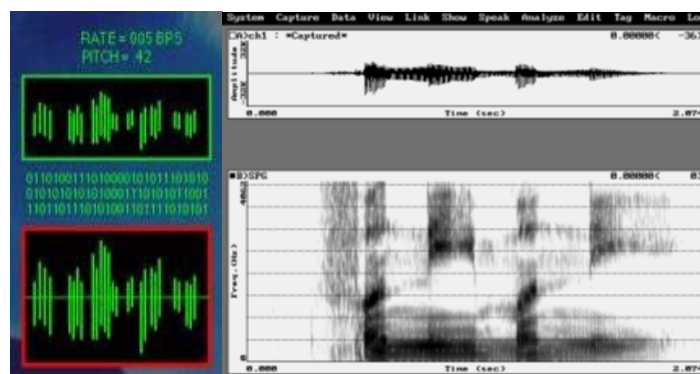


Figure 35: Voice Print Spectrum

Source: <https://developer.nvidia.com/blog/how-to-build-domain-specific-automatic-speech-recognition-models-on-gpus/>

Third step is the Verification Process stored in a table of numbers, where the presence of each dominant frequency in each segment is expressed as a binary number 0 and 1.

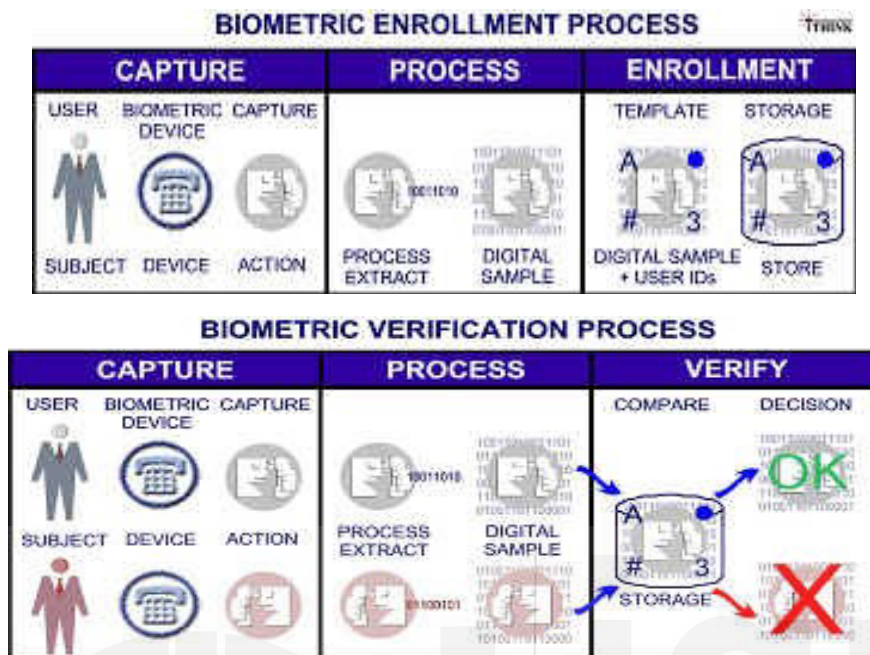


Figure 36: Biometric Process

Source: <https://www.slideshare.net/kingsingh33/biometrics-50728903>

Benefits of Voice Recognition Systems

Another layer of security for computer systems which are more efficient and cost effective adding up no ID access cards thus saves time.

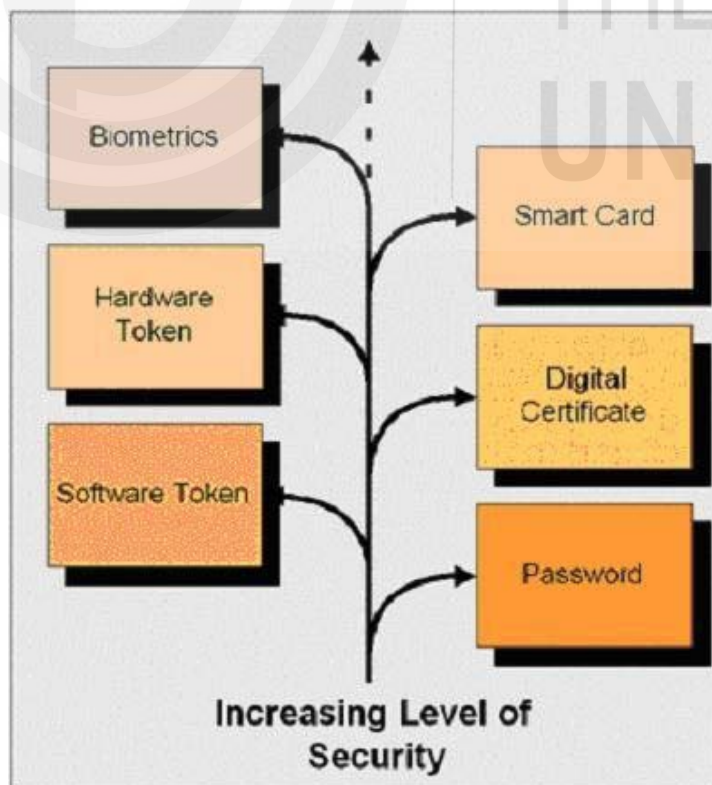


Figure 37: Increased Security Level

Source: <https://www.slideshare.net/kingsingh33/biometrics-50728903>

Drawbacks of Voice Recognition Biometrics

- ✚ Voice changes in the life time
- ✚ Uses of microphones along with environmental noise
- ✚ Poor phone, or cellular connections leading to Vocal variations: accents
- ✚ Voice samples are converted and transferred from an analog format to a digital format for processing by telephone devices.

Current Applications of Voice Biometrics

- ✚ Security system for physical access entry into a building, in telephone banking ATM machines and Voice verification in credit cards.
- ✚ Corrections industries: inmates on parole.
- ✚ Telephone companies use voice-enabling services.

The security provided by voice recognition is main benefit and further development can be make it most successful and largest applications of biometrics in the future to come.

Check Your Progress 6

- 9) Voice Recognition is becoming popular. Explain.

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4.2.6.1 Handwriting

Handwriting detection is a technique operated by Computers with the processes to receive and interpret intelligible handwritten input from sources e.g. paper documents, touch screen, photo graphs etc.

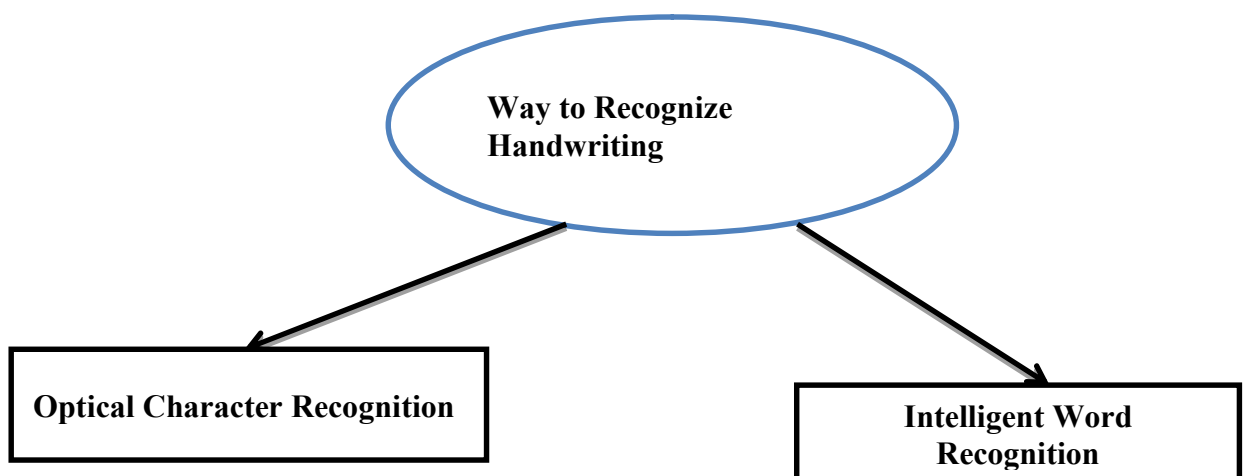


Fig. 38 Handwriting Detection

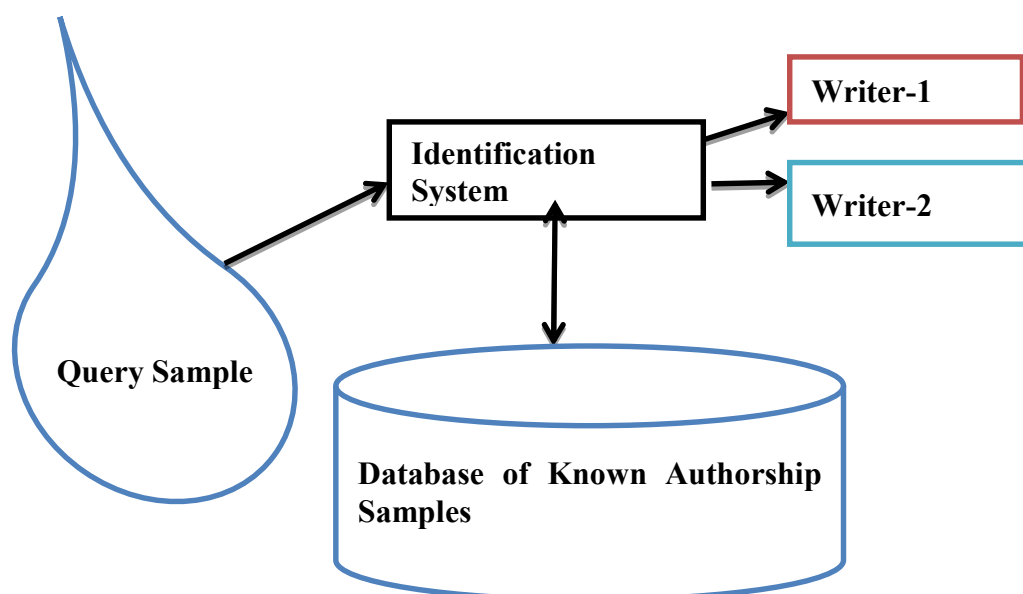


Figure 39: Working of Handwriting System (by author)

Two Types Recognition

Online Recognition

Offline Recognition

4.2.6.2 Online Recognition

On-line handwriting recognition is based on the automatic conversion of text which is written on a special digitiser and the sensor with it picks up the pen up and down movements known as digital ink, thus converted to letter codes.

The elements of the online recognition are:

- User can write with stylus or pen.
- A touch sensitive surface integrated with, or adjacent to, an output display.
- The interpretation of the movements of the stylus across the writing surface or translation of strokes into digital text is done by software.

4.2.6.3 Off-Line Recognition

Off-line handwriting recognition consists of the automatic conversion of text in an image into letter codes by the computers for text-processing applications regarded as static representation of handwriting. The different styles of writing make offline process difficult. Recognition strategies are based upon the type of data and writing and first step is Character Recognition which is based on following criteria:

Pre-processing Methods on the data and includes the use of global transforms (correlation, Fourier descriptors, etc.), local comparison (local density, intersections with straight lines, variable masks, etc.); geometrical or topological characteristics

The type of the decision algorithm relies on various statistical methods, neural networks, and structural matching

Two strategies are employed for the word recognition:

- 1) **The Holistic Approach** - recognition is globally performed on the whole representation of words without segmentation thus an advantage of the method.
- 2) **The Analytical Approach** - deals with several levels of representation i.e. words are understood as sequences of smaller size units and easily related to characters in order to make recognition independent.

Handwriting Recognition objective is to design systems for recognition of handwriting of natural language. Methods and recognition rates depend on the level of constraints on handwriting and characterised by the:

- ✚ types of handwriting
- ✚ number of scriptors
- ✚ size of the vocabulary
- ✚ spatial layout

4.2.6.4 Advantages and Disadvantages of Handwriting Recognition

Advantages:

Offline handwriting recognition has varied benefits like reading postal addresses, bank check amounts, and forms. It forms the basis for digital libraries, allowing the entry of image textual information into computers by digitization, image restoration, and recognition methods.

Disadvantages

The disadvantage of Handwriting is that information related to data or about the type of the input cannot be obtained. First, the text has to be separated into characters or words to match these words in a sequence of data by usage of algorithms.

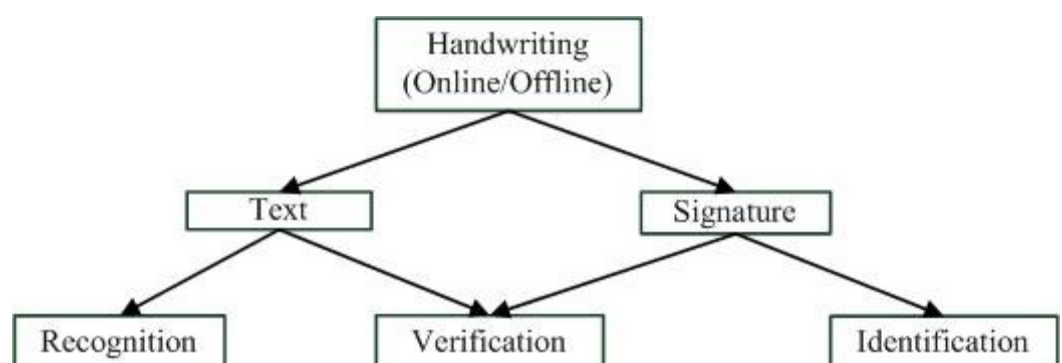


Figure 40: Handwriting Recognition Steps

Source: https://www.researchgate.net/publication/350286160_A_Review_of_Various_Handwriting_Recognition_Methods

Handwriting methodologies are difficult to operate but generate accurate data.

Check Your Progress 7

10) Describe offline and online handwriting systems.

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4.3 SUMMARY

Biometrics helps to establish a person's identity on the basis of personal traits by employing information technology techniques with accuracy. Biometrics is a technique employed to measure, compare, analyse and identify the person in terms of characteristics features like fingerprints, face, retina etc. The personal characteristic attributes like hand geometry, colour of iris, or behavioural characteristics etc. make every individual different thus biometrically scan, compare and identify with following three types modalities:

- Physiological
- Behavioural
- Both physiological and behaviour

The authentication, verification, security and accuracy has led many applications of Biometrics in technology world like:

- Workplace access control
- Identification citizenship and immigration systems of the persons
- Secrecy and confidentiality of the information
- Forensic application in criminal Identification
- E-commerce secure transactions
- Reduction in fraud and theft
- Enforcement of law

4.4 REREFERNCES

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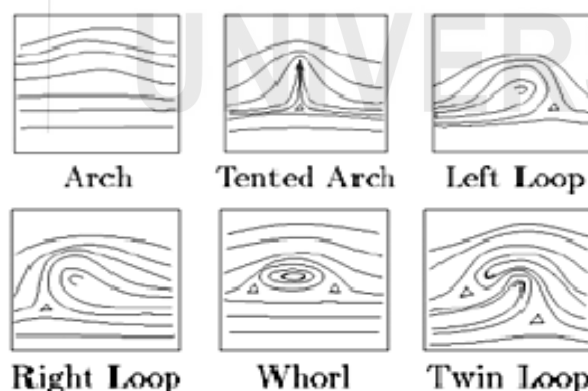
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4.5 HINTS/ ANSWERS TO CHECK YOUR PROGRESS

- 1) Biometrics is defined as the applicable means of identifying and authenticating individuals in a reliable and fast way through the use of unique biological characteristics and technologies that lead to the acceptable way of solving the crime. The identification of persons based on unique and definite characteristics leads to its application in forensic science. Refer to section 4.1.3 for detail.
- 2) The process of comparison between the known fingerprint against another in order to determine that it is of the same finger or palm is known as fingerprint recognition. Refer to section 4.2.1.4 for other parts of answer.
- 3) Finger Print Biometrics is defined as the method of identifying finger print by employing the software on the basis of pattern of finger and location of the minutiae. There are 6 patterns of fingerprints-Arch, Tented Arch, Left Loop, Right Loop, Whorl and Twin Loop.



- 4) This is an important step as image acquired would lead to recognition and the characteristics required are: Good contrast, and high quality image and high illumination. Refer to section 4.2.2.5.
- 5) Iris is made up of pigmented elastic tissue with an adjustable circular opening in the centre. Refer to section 4.2.2, Figure 14.
- 6) Retinal recognition requires that the end user must be in close proximity to retinal scanning device. The process has following steps:

- Image/Signal Acquisition and Processing:
- Enrolments and Verification Template Creation.

Refer to section 4.2.3.1 for details.

- 7) Procedure of DNA fingerprinting involves the main steps of- Collection and isolation of DNA, Cutting, size and sorting, Separation of fragments, Southern blotting and hybridisation, and DNA Fingerprint. Refer to section 4.2.5.1, Figure 31.
- 8) The double helical structure of the DNA present on saliva, hair follicles, bones, blood and sperm serve as a useful tool in the forensic studies. This can be done as the DNA from an individual's tissue shows the same degree of polymorphism. These polymorphic characters are inheritable from parents to their children.
- 9) Benefits of voice recognition systems:
 - Another layer of security for computer systems which are more efficient and cost effective adding up no ID access cards thus saves time.
 - Security system for physical access entry into a building, in telephone banking, ATM machines and voice verification in credit cards.
 - Correction industries: inmates on parole.
 - Telephone companies use voice-enabling services.
 - The security provided by voice recognition is the main benefit and further development can make it most successful and largest application of biometrics in future.
- 10) On-line handwriting recognition is based on the automatic conversion of text which is written on a special digitiser and the sensor with it picks up the pen up and down movements known as digital ink, thus converted to letter codes.

Off-line handwriting recognition consists of the automatic conversion of text in an image into letter codes by the computers for text-processing applications regarded as static representation of handwriting.